Target-Specific Recoverability Maps for Reduced-Lead ECG Reconstruction

A graded per-lead identifiability + conditioning certificate with calibrated intervals, and an honest account of an abandoned fabrication claim

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Abstract

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is ill-posed, and a scalar reconstruction error says nothing about *which* named feature's low-rank (dipolar) component, on *which* lead, is even recoverable. We make this a target-specific, computable certificate. For a waveform segment s (P/QRS/ST/T), an observed lead set S, and a target lead ℓ, two closed-form numbers from one truncated SVD of the segment's empirical rank-3 spatial submatrix give a graded *identifiability* $\eta_{s,\ell}(S)$ (its exact value zero iff the target's dipolar component is recoverable from S) and a *conditioning* $\kappa_{s,\ell}(S)$. To the map we attach an exact per-lead *minimax ambiguity floor* $a_{s,\ell}(S)$—the exact minimax risk of estimating the target's dipolar functional over a second-moment class, attained by a Gaussian least-favourable prior—which on real data *predicts* the measured per-lead dipolar error of the reconstructors we evaluate (Spearman 0.81, AUC 0.87); it is a worst-case-prior floor, so estimators exploiting predictable non-Gaussian structure fall below it, and that gap is exactly the residual our distribution-free calibrated intervals cover. On PTB-XL, precordial ST is *exactly* non-identifiable from limb leads (rank-2 frontal plane), a verdict robust to delineator and sampling rate; the total ST-threshold-event error rate's point estimates range 51.1–53.7% across the evaluated reconstructors (dominated by missed crossings) while the false-positive/negative split is a reconstructor choice. Cross-cohort (PTB-XL→Chapman) the QRS dipolar subspace is nearly shared while ST is not—an empirical sensitivity, not a certified transfer guarantee. We document in full a claim we abandoned—that a diffusion reconstructor demonstrates "fabrication is a property of the objective"—which a leakage-corrected, multi-seed re-analysis dissolved; an exploratory null-space energy descriptor and a greedy lead-selection heuristic are reported as such. Every number is regenerated from lineage-stamped artifacts.

1 Introduction

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads [1, 2]. A low aggregate error certifies nothing about *which* morphological features are trustworthy: a cardiologist reads the P wave, the QRS complex, the ST segment and the T wave, each on specific leads, and a reconstruction that gets the mean right but invents an ST deviation invents a finding.

We ask a sharper, reconstructor-independent question: *for a given observed lead set, which target lead's low-rank morphology is even identifiable, and how well-conditioned is its recovery?* Each waveform segment's instantaneous potential is approximately low rank across leads; we estimate a per-segment rank-3 spatial basis $\boldsymbol{M}_s$ (top-3 singular vectors of the population lead covariance). We stress that $\boldsymbol{M}_s$ is a *population-estimated (PCA)* object: on real PTB-XL its three principal angles to the classical vectorcardiographic (inverse-Dower) space are graded—one close (2–6°), one moderate (10–15°), and one substantially different (43–55°)—so we call it an empirical rank-3 subspace rather than a physical dipole (Sec. 5).

Contributions. Beyond the per-lead certificate (Sec. 3, proof App. A), this extended version of a four-page conference submission adds: (1) an exact per-lead *minimax ambiguity floor*—the exact minimax risk of the target's dipolar functional over a second-moment class—which, on real data, *predicts* the measured per-lead dipolar error of the evaluated reconstructors (Prop. 3, Fig. 1); (2) delineator/sampling-rate robustness of the exact ST verdict (Sec. 8); (3) an empirical cross-cohort sensitivity study (PTB-XL→Chapman) with a $\sin\Theta$ diagnostic of dipolar-subspace agreement (Sec. 6); and (4) a complete account of a claim we abandoned during pre-submission review, together with two *exploratory* descriptors—a null-space energy ratio and a greedy lead-selection heuristic—reported as exploratory, not certified (Sec. 7, Sec. 5). We report the abandonment in detail on purpose: hiding the earlier, more exciting numbers would be precisely the selective reporting our certificate is designed to expose.

2 Related Work

Lead reconstruction and VCG synthesis. Fixed linear transforms synthesize the vectorcardiogram or missing leads from a standard set (inverse-Dower [3], Kors regression [4]), and reduced-lead reconstruction has a long clinical literature [5]; recent deep and arbitrary-mask waveform-completion models push reconstruction accuracy [1, 2, 6, 7]. These target the reconstruction; we target the *prior* question of what a given configuration can support, independent of the reconstructor. Accordingly our neural baseline is a *representative* masked-completion U-Net of this family, included to show the certificate's verdict still *predicts* a flexible learned reconstructor's error—not a claim to beat [6, 7]. **Identifiability and lead selection.** Whether a linear functional is recoverable from a sub-observation is the classical estimability question [8]; choosing which leads to observe is optimal sensor selection [9]. Our per-lead η is estimability specialized to the ECG dipolar functional; the novelty is not the identity but the graded, empirically-estimated, per-target-lead map with calibrated per-record error attached. **Uncertainty for inverse problems.** Conformal prediction [10, 11] gives distribution-free intervals under exchangeability; we apply Mondrian CQR per (S, s, ℓ) group.

3 The Recoverability Certificate

Model. Within segment s write the population 12-lead vector as $\boldsymbol{L}_s = \boldsymbol{\mu}_s + \boldsymbol{M}_s\boldsymbol{d} + \boldsymbol{r}_s$, where $\boldsymbol{M}_s \in \mathbb{R}^{12\times 3}$ is the (population-estimated, PCA) dipolar basis, $\boldsymbol{d}$ the dipole coordinates, and $\boldsymbol{r}_s$ the non-dipolar residual. Observing S gives $\boldsymbol{y}_S = \mathbf{Sel}_S\boldsymbol{L}_s + \boldsymbol{n}$, with $\boldsymbol{M}_{s,S} = \mathbf{Sel}_S\boldsymbol{M}_s$. Write the exact Moore–Penrose pseudo-inverse $\boldsymbol{M}_{s,S}^{+}$ and the exact row-space projector $\boldsymbol{P}_{\mathrm{obs}} = \boldsymbol{M}_{s,S}^{+}\boldsymbol{M}_{s,S}$. For numerical deployment we also use the *rank-r truncation* at relative tolerance ϱ, $\boldsymbol{M}_{s,S}^{(\varrho)} = \boldsymbol{U}_r\boldsymbol{\Sigma}_r\boldsymbol{V}_r^{\top}$ keeping singular triplets with $\sigma_i > \varrho\,\sigma_{\max}$, with pseudo-inverse $\boldsymbol{M}_{s,S}^{(\varrho)+}$ and projector $\boldsymbol{P}_{\mathrm{obs}}^{(\varrho)} = \boldsymbol{V}_r\boldsymbol{V}_r^{\top}$.

Proposition 1 (Exact per-target-lead dipolar identifiability and conditioning)**.** *For each target lead ℓ define $\eta_{s,\ell}(S) = \left\|\boldsymbol{e}_\ell^{\top}\boldsymbol{M}_s(\boldsymbol{I} - \boldsymbol{P}_{\mathrm{obs}})\right\|_2$ and $\kappa_{s,\ell}(S) = \left\|\boldsymbol{e}_\ell^{\top}\boldsymbol{M}_s\boldsymbol{M}_{s,S}^{+}\right\|_2$ from the* exact $\boldsymbol{M}_{s,S}$*.*

(i) Identifiability. *$\eta_{s,\ell}(S) = 0$ iff $\boldsymbol{e}_\ell^{\top}\boldsymbol{M}_s$ lies in the exact row space of $\boldsymbol{M}_{s,S}$; then $\boldsymbol{e}_\ell^{\top}\boldsymbol{M}_s\boldsymbol{d}$ is a deterministic function of $\boldsymbol{M}_{s,S}\boldsymbol{d}$—two dipoles with the same observation have the same lead-ℓ dipolar value, at any SNR. If $\eta_{s,\ell}(S) > 0$ there is a $\boldsymbol{\delta} \in \ker\boldsymbol{M}_{s,S}$ (an unobserved dipole*

direction) with $\boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{\delta} \neq 0$; then $\boldsymbol{d}$ and $\boldsymbol{d} + t\boldsymbol{\delta}$ produce identical observations for all t while lead ℓ's dipolar value changes, so it is not identifiable at any SNR.

(ii) Conditioning. *On the identifiable part, the lead-ℓ error from observation noise and observed non-dipolar residual obeys* $\left|\boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{M}_{s,S}^+(\mathbf{Sel}_S \boldsymbol{r}_s + \boldsymbol{n})\right| \leq \kappa_{s,\ell}(S)\left(\|\mathbf{Sel}_S \boldsymbol{r}_s\|_2 + \|\boldsymbol{n}\|_2\right)$.

The global $\kappa_s(S) = \|\boldsymbol{M}_s \boldsymbol{M}_{s,S}^+\|_2$ (spectral norm) upper-bounds every per-lead $\kappa_{s,\ell}(S)$ (a row norm never exceeds the spectral norm; it is not their maximum).

Exact vs. ϱ-identifiability (numerical deployment). The deployed map computes $\eta_{s,\ell}^{(\varrho)}, \kappa_{s,\ell}^{(\varrho)}$ from the truncation $\boldsymbol{P}_{\text{obs}}^{(\varrho)}$, so a singular direction observed only through a value below $\varrho\,\sigma_{\max}$ is treated as unobserved. Then $\eta_{s,\ell}^{(\varrho)} > 0$ means lead ℓ is *not stably recoverable at resolution* ϱ (outside the retained numerical row space); it does *not* by itself imply the exact matrix has a null direction—a tiny-but-nonzero singular value is exactly identifiable ($\eta = 0$) yet ϱ-unidentifiable ($\eta^{(\varrho)} > 0$), as we verify numerically (`tests/test_exact_vs_rho.py`). We therefore call the $\eta^{(\varrho)}$ heatmap the *ϱ-identifiability* map; for well-separated configurations the exact and ϱ verdicts coincide, and near-rank-deficient S are reported with a ϱ-sweep and record-bootstrap CIs. **The limb-6 flagship is exact:** the six limb leads are exact linear combinations of I and II (Einthoven/Goldberger), so $\boldsymbol{M}_{s,S}$ has exact rank 2 and precordial ST is *exactly* non-identifiable from limb leads at any SNR, independent of ϱ. This exactness is conditional on the a-priori rank-3 dipole model for $\boldsymbol{M}_s$: at retained rank 2 the limb-6 observation spans all of $\boldsymbol{M}_s$ and the precordial verdict collapses to η=0; the non-identifiability appears at rank ≥3 (the third dipolar direction), so "exact" means exact *given* rank 3.
Relation to classical estimability. Part (i) is the classical estimability criterion (Gauss–Markov / [8]) for the linear functional $\boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{d}$; we claim no novelty for this identity. The contribution (Sec. 5) is its use as a graded, empirically-estimated recoverability map with per-record calibrated error rates (Sec. 4).
The non-dipolar residual (no independence claim on real data). Let $\boldsymbol{m}_s(\boldsymbol{y}_S) = \mathbb{E}[\boldsymbol{r}_s \mid \boldsymbol{y}_S]$ and $\boldsymbol{u}_s = \boldsymbol{r}_s - \boldsymbol{m}_s(\boldsymbol{y}_S)$, so $\mathbb{E}[\boldsymbol{u}_s \mid \boldsymbol{y}_S] = \mathbf{0}$. We do *not* assume $\boldsymbol{u}_s \perp \boldsymbol{y}_S$. Under squared loss the Bayes-optimal reconstruction error of the (vector) residual is

$$\inf_{\hat{\boldsymbol{r}}} \ \mathbb{E}\left\|\boldsymbol{r}_s - \hat{\boldsymbol{r}}(\boldsymbol{y}_S)\right\|_2^2 = \mathbb{E}\left[\operatorname{tr}\operatorname{Cov}(\boldsymbol{r}_s \mid \boldsymbol{y}_S)\right].$$

Part of $\boldsymbol{r}_s$ (the *empirically predictable residual*) is recovered by a predictor trained on $\boldsymbol{y}_S$ and metered with distribution-free calibrated intervals; the remainder is an *achievability gap*, established by a held-out achievability analysis for a stated predictor family—*not* declared unrecoverable a priori.

Proposition 2 (Strict independence limit – synthetic model only)**.** *If the data are generated so that a component $\boldsymbol{u}$ is statistically independent of $\boldsymbol{y}_S$ (an explicit synthetic construction), then $p(\boldsymbol{u} \mid \boldsymbol{y}_S) = p(\boldsymbol{u})$, the Bayes estimate of $\boldsymbol{u}$ under squared loss is its prior mean, and $\mathbb{E}\|\widehat{\boldsymbol{u}} - \boldsymbol{u}\|_2^2 \geq \operatorname{tr}\operatorname{Cov}(\boldsymbol{u})$ for any estimator. This strict limit is invoked* only *for the synthetic model; on real ECG we make no such independence claim.*

An exact minimax recoverability floor

Proposition 1 says *whether* a target lead's dipolar coordinate is recoverable (η=0) and *how well-conditioned* the estimate is (κ). We now give the *exact* minimax risk of estimating the target's dipolar functional from the exact (noise-free) observation, over a second-moment class of dipole priors—turning the binary verdict into a per-lead risk in physical units. Fix segment s and observed set S; let $\boldsymbol{d} \in \mathbb{R}^3$ be the dipole coordinate, $\boldsymbol{y}_S = \boldsymbol{M}_{s,S}\boldsymbol{d}$ the exact observation, and $\theta_\ell = \boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{d}$ the target dipolar functional of lead ℓ. Write the (orthogonal, symmetric) observed row-space projector $\boldsymbol{P} = \boldsymbol{M}_{s,S}^+ \boldsymbol{M}_{s,S}$, $\boldsymbol{Q} = \boldsymbol{I} - \boldsymbol{P}$, and

$$a_{s,\ell}(S)^2 = \boldsymbol{e}_\ell^\top \boldsymbol{M}_s \, \boldsymbol{\Sigma}_{Q|P} \, \boldsymbol{M}_s^\top \boldsymbol{e}_\ell, \qquad \boldsymbol{\Sigma}_{Q|P} = \boldsymbol{Q}\boldsymbol{\Sigma}_d\boldsymbol{Q} - \boldsymbol{Q}\boldsymbol{\Sigma}_d\boldsymbol{P}(\boldsymbol{P}\boldsymbol{\Sigma}_d\boldsymbol{P})^+\boldsymbol{P}\boldsymbol{\Sigma}_d\boldsymbol{Q},$$

where $\boldsymbol{\Sigma}_d \succ \boldsymbol{0}$ is a fixed dipole second-moment bound.

Proposition 3 (Exact minimax dipolar floor)**.** *Let* $\mathcal{P}(\boldsymbol{\Sigma}_d) = \{\pi : \ \mathbb{E}_\pi[\boldsymbol{d}] = \boldsymbol{0}, \ \mathbb{E}_\pi[\boldsymbol{d}\boldsymbol{d}^\top] \preceq \boldsymbol{\Sigma}_d\}$. *For the exact observation* $\boldsymbol{y}_S = \boldsymbol{M}_{s,S}\boldsymbol{d}$*, the minimax risk over* all *measurable (possibly nonlinear) estimators* $\hat\theta_\ell(\boldsymbol{y}_S)$ *of* θ_ℓ *satisfies*

$$\inf_{\hat\theta_\ell} \sup_{\pi \in \mathcal{P}(\boldsymbol{\Sigma}_d)} \mathbb{E}_\pi\big[(\hat\theta_\ell(\boldsymbol{y}_S) - \theta_\ell)^2\big] \ = \ a_{s,\ell}(S)^2,$$

attained by the Gaussian least-favourable prior $\pi^\star = \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_d)$ *and a* linear *estimator. Consequently* $a_{s,\ell}(S) = 0$ *iff* $\eta_{s,\ell}(S) = 0$*.*

Proof. **Upper bound.** For a coefficient $\boldsymbol{c}$ the linear estimator $\boldsymbol{c}^\top\boldsymbol{y}_S$ has error $(\boldsymbol{c}^\top\boldsymbol{M}_{s,S} - \boldsymbol{e}_\ell^\top\boldsymbol{M}_s)\boldsymbol{d} = \boldsymbol{g}_{\boldsymbol{c}}^\top\boldsymbol{d}$, so for every $\pi \in \mathcal{P}(\boldsymbol{\Sigma}_d)$, $\mathbb{E}_\pi[(\boldsymbol{c}^\top\boldsymbol{y}_S - \theta_\ell)^2] = \boldsymbol{g}_{\boldsymbol{c}}^\top\mathbb{E}_\pi[\boldsymbol{d}\boldsymbol{d}^\top]\boldsymbol{g}_{\boldsymbol{c}} \le \boldsymbol{g}_{\boldsymbol{c}}^\top\boldsymbol{\Sigma}_d\boldsymbol{g}_{\boldsymbol{c}}$, with equality at $\pi^\star$. As $\boldsymbol{c}$ varies, $\boldsymbol{M}_{s,S}^\top\boldsymbol{c}$ ranges over range $\boldsymbol{P}$, so $\min_{\boldsymbol{c}} \boldsymbol{g}_{\boldsymbol{c}}^\top\boldsymbol{\Sigma}_d\boldsymbol{g}_{\boldsymbol{c}}$ is the $\boldsymbol{\Sigma}_d$-projection residual of $\boldsymbol{w} := \boldsymbol{M}_s^\top\boldsymbol{e}_\ell$ onto range $\boldsymbol{P}$, $= \boldsymbol{w}^\top[\boldsymbol{\Sigma}_d - \boldsymbol{\Sigma}_d\boldsymbol{P}(\boldsymbol{P}\boldsymbol{\Sigma}_d\boldsymbol{P})^+\boldsymbol{P}\boldsymbol{\Sigma}_d]\boldsymbol{w}$. Since conditioning on $\boldsymbol{P}\boldsymbol{d}$ removes all $\boldsymbol{P}$-variance, $\boldsymbol{\Sigma}_d - \boldsymbol{\Sigma}_d\boldsymbol{P}(\boldsymbol{P}\boldsymbol{\Sigma}_d\boldsymbol{P})^+\boldsymbol{P}\boldsymbol{\Sigma}_d = \boldsymbol{\Sigma}_{Q|P}$, so this equals $a_{s,\ell}(S)^2$. Hence a linear estimator has worst-case risk $a_{s,\ell}(S)^2$ and the minimax risk is $\le a_{s,\ell}(S)^2$. **Lower bound.** The minimax risk is at least the Bayes risk under any single prior in the class; take $\pi^\star$. Its Bayes risk—the infimum over *all* estimators of $\mathbb{E}_{\pi^\star}[(\hat\theta_\ell - \theta_\ell)^2]$—is attained by the posterior mean and equals $\mathrm{Var}_{\pi^\star}(\theta_\ell \mid \boldsymbol{y}_S) = a_{s,\ell}(S)^2$ (Gaussian conditioning: $\boldsymbol{y}_S$ fixes $\boldsymbol{P}\boldsymbol{d}$ and the residual variance of $\boldsymbol{e}_\ell^\top\boldsymbol{M}_s\boldsymbol{Q}\boldsymbol{d}$ is the Schur complement $\boldsymbol{\Sigma}_{Q|P}$). Thus the minimax risk is $\ge a_{s,\ell}(S)^2$, and the two bounds coincide. $a{=}0 \iff \eta{=}0$**.** $\eta_{s,\ell}(S) = \|\boldsymbol{e}_\ell^\top\boldsymbol{M}_s\boldsymbol{Q}\|_2$. If $\eta{=}0$ then $\boldsymbol{e}_\ell^\top\boldsymbol{M}_s\boldsymbol{Q} = \boldsymbol{0}$, so $a^2 = 0$. If $\eta{>}0$ then $\boldsymbol{v} := \boldsymbol{Q}^\top\boldsymbol{M}_s^\top\boldsymbol{e}_\ell \neq \boldsymbol{0}$; because $\boldsymbol{\Sigma}_d \succ \boldsymbol{0}$ the Schur complement $\boldsymbol{\Sigma}_{Q|P}$ is positive definite on range $\boldsymbol{Q}$, so $a^2 = \boldsymbol{v}^\top\boldsymbol{\Sigma}_{Q|P}\boldsymbol{v} > 0$. □

Scope, and what the tests check. The result is *exact* and *noise-free*: the minimax risk of the dipolar functional θ_ℓ over the second-moment class $\mathcal{P}(\boldsymbol{\Sigma}_d)$ under an exact observation, with the infimum over all measurable estimators. We deliberately do *not* state a noise-inflated or finite-sample "floor." The observed-noise amplification $\kappa_{s,\ell}(S) = \|\boldsymbol{e}_\ell^\top\boldsymbol{M}_s\boldsymbol{M}_{s,S}^+\|_2$ (Prop. 1) is an *upper* bound on how observed noise/residual propagates into the estimate, not a lower risk floor; and a bootstrap minimum of $a_{s,\ell}$ over the estimated $\widehat{\boldsymbol{M}}_s$ is not a certified uniform confidence bound—so neither is claimed as part of the theorem. The certified object is the *dipolar* functional; a floor on the full lead would require bounding the non-dipolar residual, which we instead meter with the distribution-free intervals of Sec. 4. `tests/test_floor.py` numerically checks that *selected* estimators (the posterior mean, ridge, OLS) do not beat $a_{s,\ell}$ under $\pi^\star$: this validates the theorem on those estimators, it does not re-prove it.

Operational value: $a_{s,\ell}$ predicts error. The floor is a *worst-case-prior* quantity, not an empirical envelope. On real ECG its value is as a *predictor*: $a_{s,\ell}$ strongly predicts and ranks the measured per-lead dipolar error (Spearman 0.81, Pearson 0.93; Fig. 1) and the identifiable/unidentifiable split is separated with AUC 0.87. Because it is worst-case over $\mathcal{P}(\boldsymbol{\Sigma}_d)$, estimators exploiting the true (non-Gaussian) population structure fall *below* it on a fraction of the 84 non-trivial ($\eta{>}0$) cells (the pure-dipolar reconstructor on 0.21, ridge/OLS on 0.60); this below-floor gap is exactly the

Tier-II predictable residual the calibrated intervals of Sec. 4 cover, and is fully consistent with the theorem, which bounds the worst case and not the risk on any single true prior. The η-split likewise predicts a trained diffusion model's error (median 0.066 vs 0.227 mV for η=0 vs η>0; `experiments/certificate_floor_diffusion.py`).

Cross-cohort sensitivity of the verdict

The certificate is computed from a *fitted* dipolar subspace $\boldsymbol{M}_s$, so it is natural to ask how the verdict moves across cohorts. One fact is exact and useful: $\eta_{s,\ell}(S) = \|\boldsymbol{e}_\ell^\top \boldsymbol{M}_s(\boldsymbol{I} - \boldsymbol{P}_{\text{obs}})\|_2$ depends on $\boldsymbol{M}_s$ *only through its column space.* Replacing $\boldsymbol{M}_s$ by $\boldsymbol{M}_s\boldsymbol{R}$ for any $\boldsymbol{R} \in O(3)$ rotates $\boldsymbol{P}_{\text{obs}}$ and $\boldsymbol{e}_\ell^\top \boldsymbol{M}_s$ together and leaves every η unchanged; hence two cohorts that share a dipolar subspace (all principal angles zero) have identical verdicts, and in general the drift is governed by how far the two subspaces separate.

A $\sin\Theta$ (Davis–Kahan [12]) perturbation makes this quantitative but, at the conditioning we observe, *vacuously so.* After optimal alignment, $|\eta^A_{s,\ell}(S) - \eta^B_{s,\ell}(S)| \le 2\sin(\theta^\star/2)\,(1 + 2\,\kappa_s)$, where $\theta^\star$ is the largest principal angle between the two cohorts' segment-s dipolar subspaces and $\kappa_s = \max(\kappa_s^A, \kappa_s^B)$ is the larger observed-set conditioning; the projector-difference step uses the Stewart–Wedin bound and assumes the observed block keeps its rank across cohorts and that the ϱ-truncated projector coincides with the exact one. On PTB-XL→Chapman (limb-6→V2) the right-hand side evaluates to 1.37 (QRS) and 32.2 (ST), *both exceeding* the trivial bound $|\Delta\eta| \le 1$ (since $\eta \in [0, 1]$). The inequality therefore **certifies nothing here**: it does not establish that the QRS verdict transfers, and it is not a pre-deployment guarantee. A non-vacuous statement would need a target-specific bound built from the aligned target-lead rows and the exact projector difference; we leave that to future work and do not claim it.

We accordingly present the cross-cohort comparison as *empirical sensitivity*, not a certified bound (`results/transfer_bound.json`). The measured principal angle and the measured verdict drift agree qualitatively: for QRS the two dipolar subspaces are nearly aligned ($\theta^\star$=13.8°; η drift 0.033), whereas for ST they are nearly orthogonal ($\theta^\star$=85.7°, CI up to 89.6°) with a larger but highly uncertain drift (the Chapman η bootstrap CI is $[0.02, 0.71]$). This is a two-cohort observation—the QRS dipolar geometry happens to be shared between these cohorts while the low-amplitude ST geometry is not, consistent with the subspace-angle ablations of Sec. 8—not a guarantee about unseen cohorts.

4 Distribution-Free Calibration

4.1 Calibrated intervals for the predictable residual

The prediction target is the *record-level segment-mean* off-dipole residual of the target lead (one scalar per record: the mean over the segment's samples). We train a gradient-boosted quantile regressor of it on the observed leads' segment means and apply Mondrian conformalized quantile regression [10] per (S, s, ℓ) group, with strict fold discipline: basis and model on folds 1–7, the regressor's `max_iter` selected by pinball loss on fold 8, conformal calibration on fold 9, and a single evaluation on fold 10 (nominal $1-\alpha$=0.90). This is a sound *application* of established conformal machinery, not a new predictor; the contribution is attaching it, per feature and lead, to the recoverability map. Across the pre-registered (S, s, ℓ) groups, fold-10 within-group coverage clusters near nominal (0.87–0.94; e.g. {I,II,V1,V3,V5} QRS/V2 0.91, record-bootstrap CI [0.89, 0.93], mean width 0.150 mV).

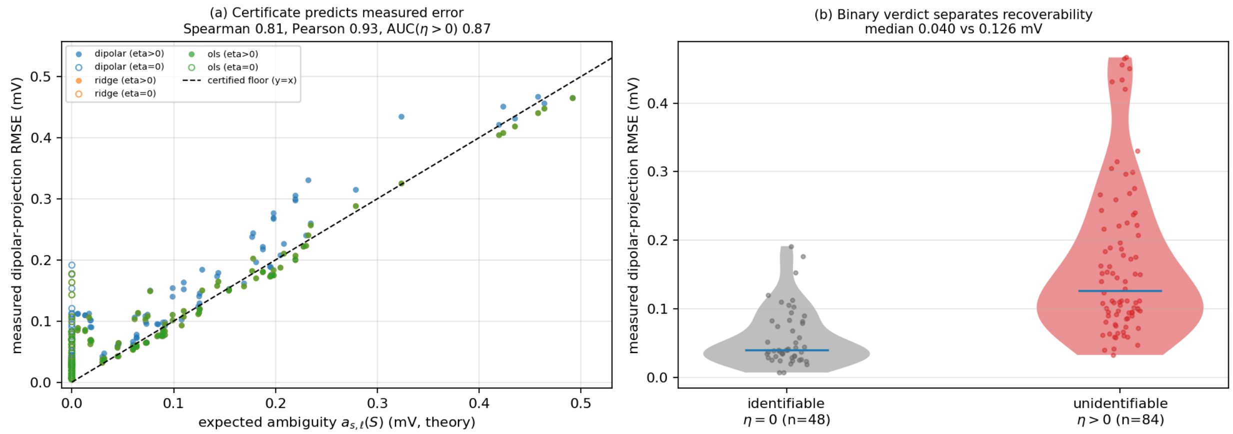

Figure 1: **The certificate is operational.** Across 132 (config, segment, target-lead) cells on held-out fold-10 records, the theoretical per-lead expected ambiguity $a_{s,\ell}(S)$ predicts the *measured* per-lead dipolar-projection error of the dipolar, ridge, and OLS reconstructors. (a) $a_{s,\ell}(S)$ strongly predicts and ranks the measured error (Spearman 0.81, Pearson 0.93). The minimax floor $a_{s,\ell}(S)$ (Prop. 3, $y=x$) is a *worst-case-prior* lower bound, not an empirical envelope: on the true (non-Gaussian) prior, estimators that exploit predictable population structure beat it on a fraction of the 84 non-trivial ($\eta>0$) cells (the pure-dipolar reconstructor on 0.21, ridge/OLS on 0.60)—exactly the Tier-II residual the calibrated intervals of Sec. 4 meter, with the below-floor gap measuring how much structure each exploits. (b) The binary verdict separates recoverability: median dipolar error 0.040 mV for identifiable ($\eta=0$) vs 0.126 mV for unidentifiable ($\eta>0$) leads (AUC 0.87). Regenerated from `results/certificate_validation.json`.

What is and is not guaranteed. Mondrian CQR gives *within-*(S, s, ℓ)*-group marginal* coverage under exchangeability—*not* coverage conditional on diagnosis, which is not a conditioning variable. Diagnostic subgroups are therefore not guaranteed and the smallest ones fall below nominal: the worst is the HYP subgroup at coverage 0.72 ($n=85$), the expected finite-sample behaviour when conditioning on a variable the calibration does not stratify on. We report worst-subgroup coverage rather than hide it under the aggregate band, and we make no distribution-free claim beyond exchangeability with the calibration fold.

5 Experiments

Data. PTB-XL [15] (21,799 twelve-lead records, 100/500 Hz, 10-fold split) with P/QRS/ST/T delineation by NeuroKit2 [16]; the map is a deterministic function of the estimated $\boldsymbol{\mu}_s, \boldsymbol{M}_s$ (record-level bootstrap CIs), not a held-out evaluation. All table numbers regenerate from result JSON via `paper/emit_baseline_table.py`.

The certificate predicts measured error. On held-out records the theoretical expected ambiguity $a_{s,\ell}$ and the measured per-lead dipolar-projection RMSE are strongly rank- and value-correlated (Spearman 0.81, Pearson 0.93; Fig. 1a), the identifiability verdict $\eta>0$ predicts a large measured error with AUC 0.87, and κ predicts the residual error *within* the identifiable regime—so the map is not a formal artifact but an operational predictor of what each reconstructor can and cannot recover.

Table 1: Limb-6 $\to$ precordial (all V1–V6) ST reconstruction, 1498 common fold-10 records. FP/FN = false-positive/false-negative ST-threshold crossings (%); total = FP+FN. FN dominates (missed crossings); the split is a reconstructor choice. Auto-generated from `results/st_safety.json`.

reconstructor	FP %	FN %	total %	|ST| err (mV)
dipolar	5.3	45.7	51.1	0.077
ridge	1.7	52.1	53.7	0.076
OLS	1.7	51.8	53.5	0.076

5.1 The recoverability map on PTB-XL

We estimate $\boldsymbol{M}_s$ (rank 3; the cardiac dipole is 3-D *a priori*, and the cumulative explained variance confirms rank 3 captures the bulk—a rank-sensitivity sweep r=2 . . . 5 is reported in `results/recoverability_maps` per segment on the training NORM records, with RECORD-LEVEL bootstrap CIs (resample records, refit $\boldsymbol{M}_s$; ϱ=10^{-2}; Fig. 2). We plot the *normalized* identifiability $\tilde{\eta}_{s,\ell} = \eta_{s,\ell}/\|\boldsymbol{e}_\ell^\top \boldsymbol{M}_s\|_2$— a dimensionless relative unobserved geometric gain (a ratio of L_2 norms, not a variance fraction). A single observed lead leaves every other lead unidentifiable; a dipole-spanning {I,II,V2} or {I,II,V1,V3,V5} makes all targets *exactly* identifiable (η=0).

The robust flagship is exact. The six limb leads are exact linear combinations of I and II, so $\boldsymbol{M}_{s,S}$ for limb-6 has *exact* rank 2: precordial ST is *exactly* non-identifiable from limb leads at any SNR, independent of ϱ (Prop. 1).

Graded ordering: stable for QRS, weighting-sensitive for ST/T. A duplicated-limb sensitivity analysis (`results/lead_weighting.json`: refit on the 8 algebraically independent leads, on the *same* records) shows the V1–V6 $\tilde{\eta}$ ordering is bootstrap-stable for QRS (Spearman 1.00, CI lower bound 0.88; anterior-before-lateral in 0.98 of resamples) but *not* robust for the low-amplitude ST/T segments (ST Spearman point 0.89 with a wide bootstrap CI; anterior-before-lateral in only 0.77 of resamples). We therefore report the ST/T $\tilde{\eta}$/ambiguity magnitudes with this caveat and *do not* claim a fine ST/T ordering; the binary limb$\to$precordial verdict above is the robust ST statement. The estimated subspace also depends on diagnosis (a NORM vs. MI principal-angle sensitivity is reported), so the map is a *NORM-trained reference*.

Downstream ST cost across reconstructors. We reconstruct the full 10-second waveform with three real continuous linear reconstructors (observed leads exact; ST at J+60 ms vs. the isoelectric PR-segment baseline [P offset − QRS onset], on fiducials from the observed Lead II shared between truth and reconstruction; median over beats; the PR-segment baseline fell back to the pre-QRS window in $\approx$ 14% of beats), and count ST-threshold events (|ST|$\geq$0.1 mV; false positive = crossing in reconstruction not truth, false negative = vice versa; Table 1). On 1498 common records the *total* wrong-event rate has point estimates ranging 51.1–53.7% across reconstructors and is *dominated by false negatives* (missed true crossings: limb$\to$precordial recovery smooths ST toward the population), with the false-positive/false-negative split a reconstructor choice (paired record-bootstrap Δ on the FP/FN split, and on the total, in `results/st_safety.json`). We do not derive a minimax/Bayes bound and make no equivalence claim: we report the range of total-error point estimates, *not* a certified lower bound, and ST-threshold events, not diagnoses. Because the PR-segment baseline fell back to the pre-QRS window in $\approx$ 14% of beats, the exact ST magnitudes inherit that delineation limitation.

Truncation-tolerance sensitivity. Well-posed configurations are ϱ-stable ({I,II,V1,V3,V5} is rank 3 across $\varrho \in \{10^{-4}, \ldots, 10^{-1}\}$); near-rank -deficient sets are ϱ-sensitive and reported with the ϱ-sweep and bootstrap CIs. A tiny-but-nonzero singular value is exactly identifiable yet ϱ-

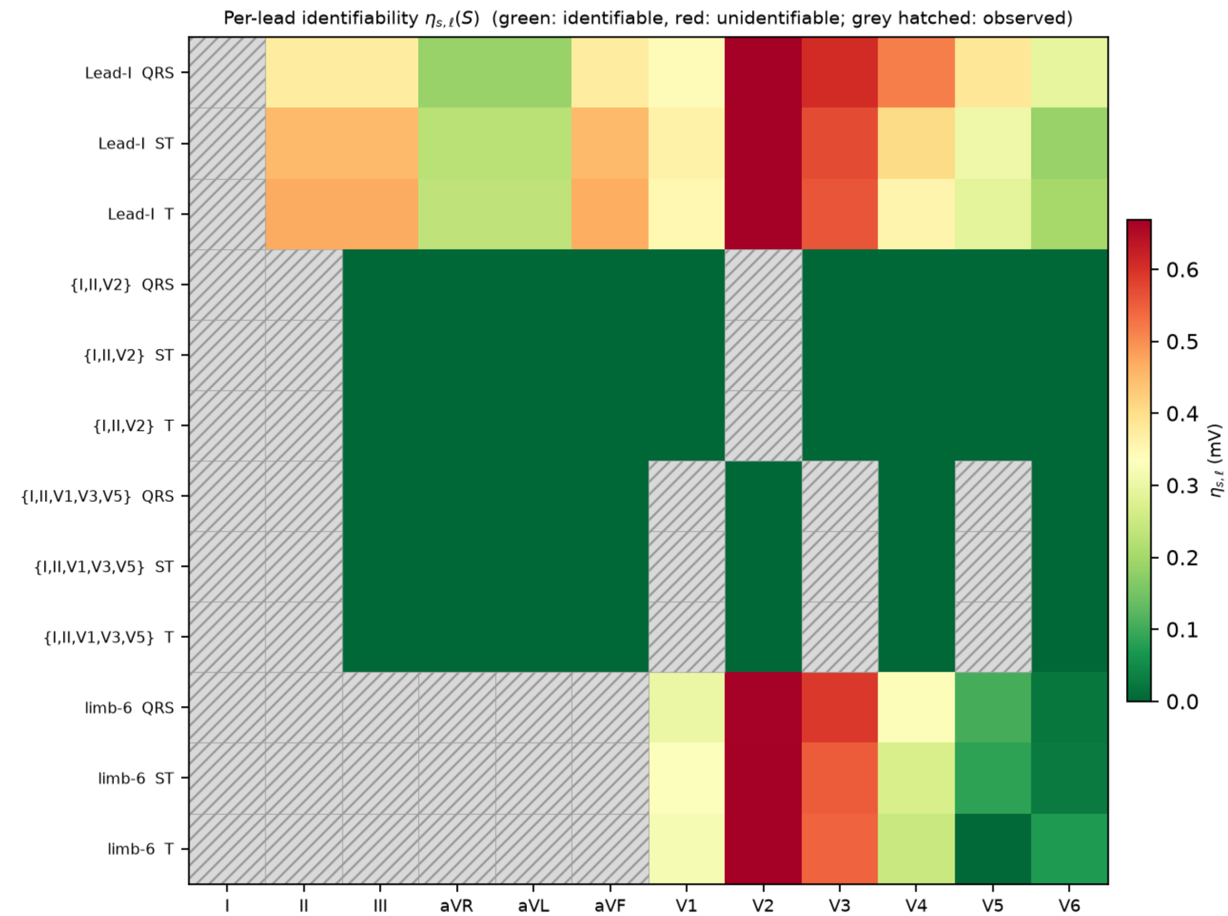

Figure 2: Recoverability map: normalized identifiability $\tilde{\eta}_{s,\ell}(S)$ (left, dimensionless) and prior-conditional expected ambiguity in mV (right), colorblind-safe `cividis`; grey hatched = observed. For limb-6 all precordial leads are unidentifiable (η>0); the fine ST/T ordering is weighting-sensitive (text).

unidentifiable, which is why we separate exact from ϱ-identifiability (Prop. 1).

5.2 Baselines and what the map predicts

Table 2 reports reconstruction RMSE (mV) of the unobserved target leads under a *single, identical protocol* for all methods: the same shared split (train folds 1–7, ridge λ selected on fold 8, test on NORM fold-10), the same 800 test records, per-timepoint waveform RMSE on the same segment indices, and *paired* record-bootstrap CIs on the step-to-step differences. We include a *representative* masked-completion baseline—an arbitrary-mask conditional 1-D U-Net trained with random lead masks (one model serves any configuration, no target-lead leakage), the same family as [1,6,7], run over three seeds (mean reported)—not a state-of-the-art claim.

On a dipole-spanning set the methods form a ladder in point estimates (the ridge→U-Net step is within its paired CI)—prior-mean → dipolar (Tier I) → per-segment ridge (Tier I+II) → U-Net—evidence of recoverable non-dipolar content (the calibrated Tier II layer), though the neural model's margin over a simple ridge is small (QRS 0.164 vs. 0.167 mV on {I,II,V1,V3,V5};

Table 2: Reconstruction RMSE (mV) of target leads, QRS segment, under one identical per-timepoint protocol on NORM fold-10 (800 test records, shared split). "dipolar" = Tier I; "ridge" = per-segment ridge (Tier I+II); "U-Net" = representative masked-completion baseline (3-seed mean). Best in bold. On limb-6 even the neural baseline plateaus far above its identifiable-set error. Numbers auto-generated from `results/fair_baselines.json`.

config	prior-mean	dipolar	ridge	U-Net
{I,II,V2}	0.467	0.236	0.220	**0.183**
{I,II,V1,V3,V5}	0.481	0.183	0.167	**0.164**
limb-6	0.481	0.361	0.344	**0.238**

paired Δ CI $[-0.007, +0.000]\,\mathrm{mV}$). On **limb-6** the U-Net still *lowers* error over the prior mean (QRS 0.481 $\rightarrow$ 0.238)—it recovers the predictable, largely population-correlated structure—but plateaus at 1.46$\times$ its spanning-set error (paired spanning$\rightarrow$limb-6 Δ CI $[+0.070, +0.079]\,\mathrm{mV}$) and no reconstructor closes that gap. This residual gap is *consistent with* the unobserved dipolar coordinate the map's $\eta{>}0$ marks as unidentifiable (we do not decompose the error into projected components, so we claim consistency, not equality): capacity recovers the predictable part but not the missing coordinate, so a reconstruction that looked confident about anterior precordial morphology here would be inventing it. We therefore scope the identifiability claim to the dipolar component, not to a blanket "not recoverable."

Empirical, not physical, subspace. On real PTB-XL the estimated $\boldsymbol{M}_s$ has three graded principal angles to the classical inverse-Dower vectorcardiographic space [3, 4]: one close (2–6°), one moderate (10–15°), and one substantially different (43–55°, bootstrap CIs excluding small angles). We therefore treat $\boldsymbol{M}_s$ as an *empirical rank-3 spatial subspace* estimated from the data, not a physical cardiac dipole; all recoverability quantities are stated with respect to this estimated subspace. This empirical, data-driven subspace—rather than a fixed physiological transform—is what makes the map dataset-specific and, we verify, stable across PTB-XL and Chapman for the QRS segment (and robust to the duplicated-limb weighting).

An exploratory prescriptive use. Because $a_{s,\ell}(S)$ is a function of the chosen leads, one can also *select* them: minimising the total ambiguity $J(S) = \sum_{s,\ell} a_{s,\ell}(S)^2$ over an *abstract lead-channel budget* k, choosing among the eight independent measured lead channels (we do *not* model physical electrode placement or Wilson-terminal constraints, so this is a channel-selection abstraction, not an electrode-placement recommendation). As an *exploratory* result on this single fitted cohort, greedy forward selection matches the exhaustive optimum at every budget, and the ambiguity reduction satisfies the diminishing-returns inequality in 1.00 of 200 randomized tests—an empirical check, not a submodularity proof, so we claim *no* approximation guarantee. Descriptively the most informative single channel is $V4$, the best pair $\{V4, V2\}$ (J=0.19 vs $0.31\,\mathrm{mV}^2$ for a random pair), and three chosen channels $\{V4, V2, II\}$ suffice to reach J=0 (they span the rank-3 dipole). Cross-cohort and bootstrap stability of the selection would be needed before any placement recommendation (`experiments/active_selection.py`).

6 Cross-Dataset Transfer (Matched Comparison)

To test whether the certificate is an artifact of one cohort we re-estimate $\boldsymbol{M}_s$ under the same unified truncated-SVD numerics on the independent Chapman-Shaoxing-Ningbo 12-lead database [17] and compare principal angles, rank, and κ to PTB-XL. To avoid confounding *cohort* with *case mix*, we run a **matched** comparison (PTB-XL NORM vs. Chapman sinus-rhythm records) alongside an all-record comparison. Chapman records are sampled *randomly* from the complete manifest with a fixed seed (not a deterministic folder/name stride), channels are reordered by WFDB `sig_name` with a 12-lead assertion and resampled from the record's actual sampling rate, and the exact loaded manifest is hashed (`results/cross_dataset.json`); we used 350 Chapman records, of which 81 were sinus-rhythm.

QRS transfers; ST/T do not. In the *matched* (normal-vs-normal) comparison the QRS subspaces are close (max principal angle 14°, record-bootstrap CI $[10, 25]$), and the well-posed conditioning transfers (κ for {I,II,V1,V3,V5}: 2.4 PTB-XL vs. 2.3 Chapman). limb-6 is effective rank 2 on both cohorts (we do not invert a near-zero singular value into a spurious 10^5-scale κ). The low-amplitude segments differ sharply even under the matched comparison (ST max angle 86°, T 81°), and the all-record comparison is similar (QRS 14°, ST 81°). Because the ST/T difference *survives the matched case mix*, we describe it as cohort-specific: the QRS map and the limb-6 verdict are cohort-stable, while the ST/T third spatial direction is cohort-specific and should be re-estimated per dataset.

Duplicated-limb weighting (`results/lead_weighting.json`). On the *same* records as the primary map, refitting on the 8 algebraically-independent leads leaves the identifiability *verdict* unchanged (spanning-set η identical; limb-6 precordial stays unidentifiable) and the QRS $\tilde{\eta}$ ordering bootstrap-stable, but the low-amplitude ST/T ordering is *not* robust to the weighting (Sec. 5); we therefore make no fine ST/T ordering claim under either fit.

7 A Negative Result: the Generative "Fabrication-Objective" Story Did Not Survive

This project began with a stronger claim—that a diffusion reconstructor would demonstrate "fabrication is a property of the objective." A pre-submission re-analysis abandoned it. We document the abandonment in full because the negative result is itself informative and because selective reporting of the earlier (favourable) numbers would be exactly the failure the certificate is meant to prevent.

Setup. An arbitrary-mask conditional DDPM (12-channel binary mask + observed values; random-mask training so one model serves any configuration, with explicit leakage tests) reconstructs {V2,V4,V6} from {I,II,V1,V3,V5} under classifier-free-style guidance weight $w \in \{1, 2, 4, 6\}$ (`results/gpu_deficit_ci.json`, `realism_metrics.json`, `gpu_oracle_gate.json`; all lineage-stamped, numbers auto-generated).

(1) The original effect was a configuration-leakage artifact. The earlier exhibit reused one model trained on a fixed observed set to score a *different* configuration. With leakage removed (arbitrary-mask model + leakage guards) and 5 training seeds, the generative reconstructor does *not* outperform a simple held-out *ridge achievability comparator* (a ridge fit on the observed leads, used as an achievability reference—*not* a Bayes-optimal oracle, since the model can and does exceed it at low guidance). The deficit $\Delta\rho = \rho_{\text{ridge}} - \rho_{\text{model}}$—how far the model's target-lead correlation falls below the ridge comparator ρ_{ridge}—is *monotone* in w: mean $\Delta\rho = -0.026, -0.006, +0.041, +0.072$ for $w = 1, 2, 4, 6$ (mean over 5 seeds; SD ± 0.044 at w=6). It is small and negative at w=1 (the model marginally beats the comparator) and grows to a modest but seed-*significant* +0.072 at w=6 (95% t-interval $[+0.011, +0.134]$, *excludes zero*; the model now falls short of the comparator). Guidance thus moves the gap the *wrong* way for the fabrication story: it does not buy achievability beyond a plain ridge, it erodes the model's small low-guidance edge and drives it significantly below the comparator. The previously reported large diffusion advantage does not reproduce.

(2) Realism does not improve with guidance—the causal premise is false. The story required higher guidance to buy realism at the cost of fidelity. Instead realism is *no better* at high guidance: from w=1 to w=6 the target-lead PSD log-distance rises (V2: $1.88 \to 2.25$; a small dip at w=2 then a clear rise, so worsening but not monotone) and the amplitude-Wasserstein distance rises (V2: $0.45 \to 0.83$). The trend is toward *worse* realism on essentially every target-lead axis (both PSD log-distance and amplitude-Wasserstein rise from w=1 to w=6); there is no monotone realism/fidelity trade to exploit. This is consistent with fundamental limits on hallucination in inverse problems [18] and the perception–distortion tradeoff [19]: raising perceptual pressure (guidance) cannot manufacture target-lead information the observed leads do not carry.

(3) What is true instead. The predictable non-dipolar residual is real but modest: ridge-comparator correlations (`gpu_oracle_gate.json`) are $\rho \approx 0.49$–0.55 for V2 (QRS/ST) and 0.21–0.34 for V4/V6—consistent with the calibrated Tier-II intervals of Sec. 4. So the honest statement is a *graded predictability* of the residual, calibrated per feature and lead, not a proof of fabrication. We removed the fabrication-objective claim and rebuilt the paper around the parts that withstand scrutiny.

(4) An exploratory null-space energy diagnostic. The certificate isolates the dipole subspace $\boldsymbol{Q} = \boldsymbol{I} - \boldsymbol{P}_{\text{obs}}$ the observed leads cannot constrain. As a *descriptive, exploratory* diagnostic we measure the fraction of a reconstruction's dipolar energy placed there—the *null-space dipolar energy ratio* $R_Q = \|\boldsymbol{M}_s\boldsymbol{Q}\hat{\boldsymbol{d}}\|^2/\|\boldsymbol{M}_s\hat{\boldsymbol{d}}\|^2$ (Fig. 3). We are explicit about what R_Q is *not*: $R_Q{>}0$ means a reconstruction asserts energy in the unidentifiable subspace, but it does *not* by itself demonstrate fabrication. The true dipole can carry $\boldsymbol{Q}$-energy, and under correlated observed/unobserved dipole coordinates the Bayes posterior mean $\mathbb{E}[\boldsymbol{Qd} \mid \boldsymbol{Pd}]$ is generally *nonzero*; hence a reconstruction with R_Q=0 (which sets $\boldsymbol{Q}\hat{\boldsymbol{d}}$=0) is *not* the posterior mean and R_Q=0 is not "correct abstention." A defensible metric would instead standardize the deviation $\boldsymbol{Q}\hat{\boldsymbol{d}} - \mathbb{E}[\boldsymbol{Qd} \mid \boldsymbol{Pd}]$ by $\boldsymbol{\Sigma}_{Q|P}$ (Sec. 3); we leave that calibrated assertion score to future work and report R_Q only as an exploratory descriptor. Descriptively: reconstructors that leave the unobserved coordinate near zero have $R_Q \approx 0$ (≤ 0.000); on a heavily under-determined single lead ridge/OLS put $R_Q = 0.50$ (Lead-I QRS) into $\boldsymbol{Q}$, and the diffusion model places most of its dipolar content there ($R_Q \approx 0.75$), *increasing* with guidance ($0.75 \to 0.84$ QRS, $\to 0.86$ ST as w:1 $\to$ 6). The rank-2 limb-6 set does *not* span the rank-3 dipole, yet $R_Q \leq 0.00$ there for these reconstructors—an empirical observation that the fitted linear/generative maps do not excite the one unobserved limb-6 direction, *not* evidence that limb-6 is dipole-spanning. We do *not* read any of this as certified fabrication.

8 Limitations

The subspace $\boldsymbol{M}_s$ is estimated, so η/κ inherit its estimation error (reported via record-bootstrap CIs and the rank/ϱ/subspace-angle sensitivities of Fig. 4) and its cohort dependence (Sec. 6); the ST/T third direction is cohort-specific. Since η depends on $\boldsymbol{M}_s$ only through its column space, cross-cohort drift is governed by the principal angle between the cohorts' dipolar subspaces (Sec. 3); empirically this angle is small for QRS (13.8°, measured drift 0.033) and near-orthogonal for ST (85.7°, uncertain drift)—an empirical two-cohort sensitivity, not a certified transfer guarantee (the $\sin\Theta$ inequality is vacuous at the observed conditioning). Identifiability is stated for the low-rank (dipolar) coordinate: a reconstructor may still recover predictable non-dipolar structure (Sec. 5), so $\eta{>}0$ bounds the dipolar component, not every waveform detail. Conformal coverage is within-(S, s, ℓ)-group marginal under exchangeability, not conditional on diagnosis; rare diagnostic subgroups are under-covered (Sec. 4). Delineation uses NeuroKit2 and inherits its errors, but the ST-safety endpoint is robust to it: across a delineator×rate grid (`dwt`/`cwt` $\times$ 100/500 Hz) the

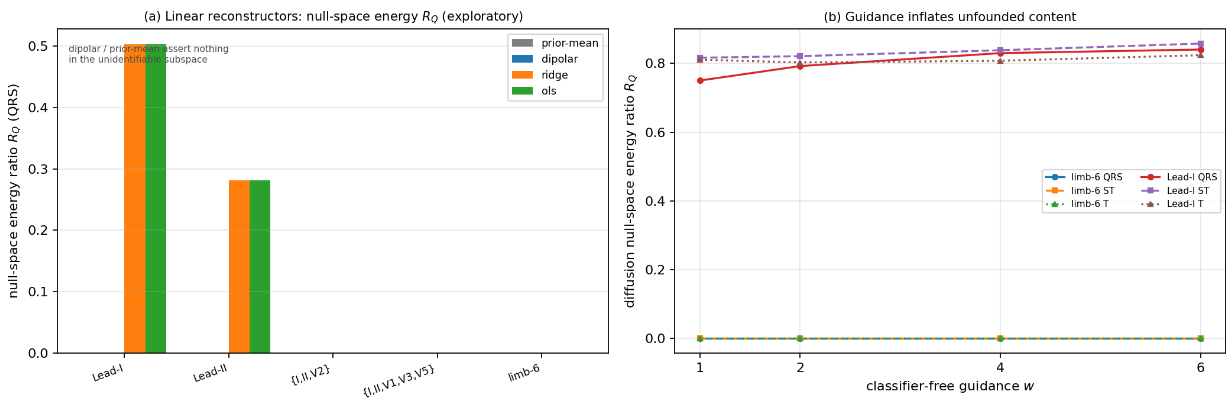

Figure 3: **Null-space dipolar energy diagnostic** (exploratory, ground-truth-free). R_Q = fraction of a reconstruction's dipolar energy in the unidentifiable subspace $\boldsymbol{Q}$; $R_Q>0$ is a descriptor, *not* proof of fabrication (Sec. 7). (a) Linear reconstructors leave the unobserved coordinate near zero ($R_Q\approx 0$) except on a heavily under-determined single lead, where ridge/OLS place energy in $\boldsymbol{Q}$. (b) The diffusion model places most of its dipolar content in $\boldsymbol{Q}$ on a single lead, increasing with classifier-free guidance w; on the rank-2 limb-6 set (which does *not* span the dipole) the fitted maps happen not to excite the one unobserved direction, so $R_Q\approx 0$. Regenerated from `results/fabrication_audit.json`, `fabrication_diffusion.json`.

precordial-not-recoverable verdict is sign-stable ($\eta>0$ at every setting, as forced by the structural rank-2 limb subspace), the per-lead recoverability ordering is preserved (Spearman ≥ 0.94), the rate axis is tight (strong-lead $|\Delta\eta| \leq 0.028$), and the endpoint stays FN-dominated (under-warning, the conservative direction) in every setting (`results/delineator_robustness.json`). The certificate is a pre-reconstruction screen, not a substitute for clinical validation.

9 Conclusion

Reduced-lead ECG reconstruction has a known low-rank structure per segment. Exploiting it gives a graded, per-target-lead recoverability map—what is identifiable, how well-conditioned, and (via calibration) what interval the predictable residual admits—before and independent of any reconstructor, plus a total ST-threshold error rate that is empirically similar across the evaluated reconstructors. It converts "trust the reconstruction" into a per-feature, per-lead, calibrated statement, and it told us honestly when one of our own earlier claims was not supported.

A Proof of Proposition 1

We work with the *exact* $\boldsymbol{M}_{s,S} = \mathbf{Sel}_S \boldsymbol{M}_s$, its exact Moore–Penrose inverse $\boldsymbol{M}_{s,S}^{+}$, and the exact orthogonal row-space projector $\boldsymbol{P}_{\mathrm{obs}} = \boldsymbol{M}_{s,S}^{+}\boldsymbol{M}_{s,S}$ (projecting onto $\mathrm{row}(\boldsymbol{M}_{s,S}) = \ker(\boldsymbol{M}_{s,S})^{\perp}$).

(i) Identifiability. The dipolar part of lead ℓ is the linear functional $f_\ell(\boldsymbol{d}) = \boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{d}$. From S we observe $\boldsymbol{M}_{s,S}\boldsymbol{d}$, which fixes $\boldsymbol{P}_{\mathrm{obs}}\boldsymbol{d}$ exactly; the exact kernel component $(\boldsymbol{I}-\boldsymbol{P}_{\mathrm{obs}})\boldsymbol{d}$ is unobserved. Decompose $\boldsymbol{e}_\ell^\top \boldsymbol{M}_s = \boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{P}_{\mathrm{obs}} + \boldsymbol{e}_\ell^\top \boldsymbol{M}_s(\boldsymbol{I} - \boldsymbol{P}_{\mathrm{obs}})$. If $\eta_{s,\ell}(S) = \|\boldsymbol{e}_\ell^\top \boldsymbol{M}_s(\boldsymbol{I} - \boldsymbol{P}_{\mathrm{obs}})\|_2 = 0$ then $f_\ell(\boldsymbol{d}) = \boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{P}_{\mathrm{obs}} \boldsymbol{d}$ depends on $\boldsymbol{d}$ only through the observed $\boldsymbol{P}_{\mathrm{obs}}\boldsymbol{d}$, hence is determined by $\boldsymbol{M}_{s,S}\boldsymbol{d}$: identifiable. Conversely if $\eta_{s,\ell}(S) > 0$ there is a unit $\boldsymbol{\delta} \in \ker \boldsymbol{M}_{s,S}$ with $\boldsymbol{e}_\ell^\top \boldsymbol{M}_s \boldsymbol{\delta} \neq 0$; then $\boldsymbol{d}$

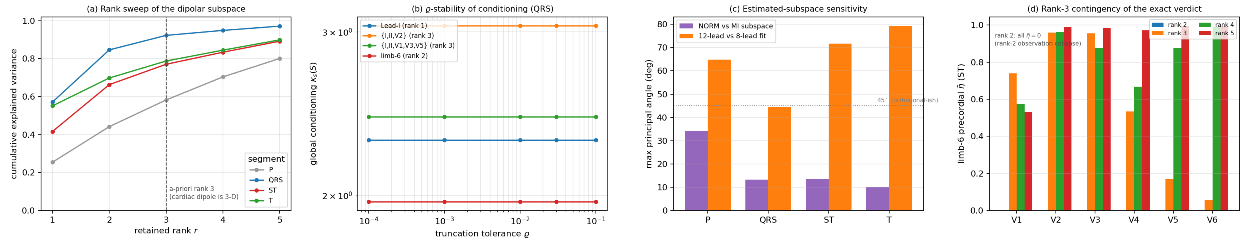

Figure 4: Sensitivities of the certificate. (a) Rank sweep: cumulative explained variance of the per-segment dipolar subspace; the *a-priori* rank-3 choice (the cardiac dipole is 3-D) captures the bulk (QRS/ST). (b) ϱ-stability: global conditioning $\kappa_s(S)$ is flat in the truncation tolerance for the deployed observed-lead sets (near-rank-deficient sets, reported in `results/recoverability_maps.json`, are ϱ-sensitive). (c) Estimated-subspace sensitivity: max principal angle per segment under a NORM→MI diagnosis shift and under the 12-vs-8-lead weighting; the low-amplitude ST/T subspaces move most (near/above 45° for the weighting), which is why we make no fine ST/T ordering claim. (d) Rank-3 contingency of the exact limb-6 verdict: at retained rank 2 the limb-6 observation is itself rank-2, so all precordial $\tilde{\eta}$=0 (a spurious "identifiable" collapse); the precordial non-identifiability appears only at rank ≥3, so the exact verdict is exact *given* the a-priori rank-3 dipole model.

and $\boldsymbol{d} + t\boldsymbol{\delta}$ give identical observations for all t but change f_ℓ by $t\,\boldsymbol{e}_\ell^\top \boldsymbol{M}_s\boldsymbol{\delta}$, so no estimator recovers f_ℓ at any SNR. This is the exact estimability criterion [8] for f_ℓ.

Exact vs. ϱ-truncated verdict (numerical check). The deployed heatmap replaces $\boldsymbol{P}_{\mathrm{obs}}$ by the rank-r truncation $\boldsymbol{P}_{\mathrm{obs}}^{(\varrho)} = \boldsymbol{V}_r\boldsymbol{V}_r^\top$ (singular directions with $\sigma_i \le \varrho\,\sigma_{\max}$ dropped). This can disagree with the exact verdict only near a truncation boundary, and only in the *conservative* direction: $\eta_{s,\ell}^{(\varrho)} \ge 0$ can be positive where $\eta_{s,\ell} = 0$ (a tiny-but-nonzero singular value is exactly identifiable yet not stably recoverable at resolution ϱ), but not vice versa. We verify all three regimes in code (`tests/test_exact_vs_rho.py`): (a) an exactly rank-deficient $\boldsymbol{M}_{s,S}$ gives $\eta = \eta^{(\varrho)} > 0$; (b) $\boldsymbol{M}_{s,S} = \mathrm{diag}(1, 1, 10^{-6})$ gives $\eta = 0$ but $\eta^{(\varrho)} > 0$ at $\varrho = 10^{-2}$; (c) a full-rank well-conditioned $\boldsymbol{M}_{s,S}$ gives $\eta = \eta^{(\varrho)} = 0$. The limb-6 flagship falls in regime (a) exactly (rank 2), so its verdict is ϱ-independent.

(ii) Conditioning. The mean-centred estimate is $\widehat{\boldsymbol{L}}_s - \boldsymbol{\mu}_s = \boldsymbol{M}_s\boldsymbol{M}_{s,S}^+(\boldsymbol{y}_S - \boldsymbol{\mu}_{s,S})$ with $\boldsymbol{y}_S - \boldsymbol{\mu}_{s,S} = \boldsymbol{M}_{s,S}\boldsymbol{d} + \mathbf{Sel}_S\boldsymbol{r}_s + \boldsymbol{n}$. On the identifiable part its lead-ℓ error from the observed non-dipolar residual and noise is $\boldsymbol{e}_\ell^\top \boldsymbol{M}_s\boldsymbol{M}_{s,S}^+(\mathbf{Sel}_S\boldsymbol{r}_s + \boldsymbol{n})$, and by Cauchy–Schwarz $|\boldsymbol{e}_\ell^\top \boldsymbol{M}_s\boldsymbol{M}_{s,S}^+(\mathbf{Sel}_S\boldsymbol{r}_s + \boldsymbol{n})| \le \|\boldsymbol{e}_\ell^\top \boldsymbol{M}_s\boldsymbol{M}_{s,S}^+\|_2(\|\mathbf{Sel}_S\boldsymbol{r}_s\|_2 + \|\boldsymbol{n}\|_2) = \kappa_{s,\ell}(S)(\|\mathbf{Sel}_S\boldsymbol{r}_s\|_2 + \|\boldsymbol{n}\|_2)$, using Cauchy–Schwarz and the triangle inequality. Each per-lead conditioning is a row norm of $\boldsymbol{A} = \boldsymbol{M}_s\boldsymbol{M}_{s,S}^+$, so $\kappa_{s,\ell}(S) = \|\boldsymbol{e}_\ell^\top \boldsymbol{A}\|_2 \le \|\boldsymbol{A}\|_2 = \kappa_s(S)$; the global κ_s upper-bounds every per-lead value but is not their maximum. □

B Reproducibility

Code, fixed seeds, an exact environment lock (`env.lock.txt`), and a one-command pipeline (`experiments/run_all` are released; every table number is regenerated from result JSON (`paper/emit_baseline_table.py`), and a CI workflow runs the test suite on every push. The neural baseline is a 24→12-channel 1-D U-Net (base width 64, residual GroupNorm blocks, 60 epochs, 3 deterministic seeds, Adam $2{\times}10^{-4}$, MSE on masked leads, random-mask training); the residual predictor is a gradient-boosted quantile regressor (pinball loss) with strict fold discipline (folds 1–7 train, 9 calibrate, 10 test).

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